Model Program Book

SHORT TERM INTERNSHIP

(Virtual)

Designed & Developed by

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF GOVERNMENT OF ANDHRA PRADESH)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

(Virtual)

Name of the Student: Thota Venkata Rao

Name of the College: Lingayas Institute of Management and Technology

Registration Number: 22NA1A6103

Period of Internship: From: 20/05/25 To: 06/07/25

Name & Address of the Intern Organization Sandspace Technologies Private Limited
Address : 2nd Floor, S B I, NCK Plaza, NTR Circle, Vijayawada, Andhra Pradesh 520010

University: Jawaharlal Nehru Technological University, Kakinada

YEAR
2025

An Internship Report on

(tittle of the internship)

Java Full Stack

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

(Name of the Faculty Guide)

Department of

Artificial Intelligence & Machine learning

Lingayas Institute of Management & Technology

Thota Venkata Rao

Reg. No: 22NA1A6103

Department of

Artificial Intelligence & Machine learning

Lingayas Institute of Management & Technology

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP StateCouncil of Higher Education **https://apsche.ap.gov.in**

- It is mandatory for all the students to complete 2 months (180 hours) of short- term internship either physically or virtually.
- Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
- Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
- You should maintain punctuality in attending the internship. Daily attendance is compulsory.
- You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
- While you are attending the internship, follow the rules and regulations of the intern organization.
- While in the intern organization, always wear your College Identity Card.
- If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
- You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
- Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and Information you are expected to collect about the organization and/or industry.
    b. Job Skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
- Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
- Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

- Never be hesitant to ask questions to make sure you fully understand whatyou need to do your work and to contribute to the organization.
- Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
- At the end of internship, you shall be evaluated by your Supervisor of theintern organization.
- There shall also be evaluation at the end of the internship by the FacultyGuide and the Principal.
- Do not meddle with the instruments/equipment you work with.
- Ensure that you do not cause any disturbance to the regular activities of the intern organization.
- Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
- You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
- If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
- Do not forget to keep up your family pride and prestige of your College.

……..<<@>>…….

Student's Declaration

I am Thota Venkata Rao a student of Bachelor of Technology Program, Reg. No: 22NA1A6103 of the Department of Artificial Intelligence & Machine learning College do hereby declare that I have completed the mandatory internship from 20/05/2025 to 06/07/2025 in Sandspace Technologies Private Limited under the Faculty Guideship of (Name of the Faculty Guide), Department of Artificial Intelligence & Machine learning, Lingayas Institute of Management and Technology.

(Signature and Date)

Official Certification

This is to certify Thota Venkata Rao that Reg. No.22NA1A6103 has completed his/her Internship in Sandspace Technologies Private Limited on Full Stack Java under my supervision as a part of partial fulfillment of the requirement for the Department of Artificial Intelligence & Machine learning, Lingayas Institute of Management and Technologies.

(Signatory with Date and Seal)

Endorsements

Faculty Guide

Head of the Department

Principal

Certificate from Intern Organization

This is to certify that Thota Venkata Rao Reg. No 22NA1A6103 of Lingayas Institute of Management & Technology underwent internship in Sandspace Technologies Private Limited from 20/05/2025 to 06/07/2025

The overall performance of the intern during his/her internship is found to be

_____ (Satisfactory/Not Satisfactory).

Authorized Signatory with Date and seal

Certificate from Intern Organization

TEGA ISO 9001:2015 nasscom

CERTIFICATE

of COMPLETION

This is to certify that Thota Venkatrao

has successfully completed an internship in the

Java Full stack at Sandspace Technologies Pvt Ltd

from May 2025 to July 2025.

We acknowledge the valuable contribution made during the internship and extend our best wishes for future achievements.

Sandspace Technologies Private Limited

HR MANAGER

Acknowledgements

I Thank our Honourable chairman sir, **Mr. Gadde Rajling** for his moral support and the excellent facilities provided.

I am grateful to_____**,** Principal, Lingayas Institute of Management & Technology, Nuzvid, for his support throughout the course and for giving the permission to undergo this internship work.

I wish to place on record our profound gratitude to **Prof. Ch. Srinivasa Rao** Head of the Department, Department of Computer Science and Engineering for having given me the opportunities to complete this internship successfully.

Firstly, I indebted to my parents, friends and relatives for their physical, psychological and moral extended by them at all time.

A special thanks to ChatGPT, Google, Gemini AI & Youtube for providing awesome facilities.

Thota Venkatarao

(22NA1A6103)

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CHAPTER 1: EXECUTIVE SUMMARY

The report epitomizes about internship performed with the time span of two months in **Sandspace Technologies Private Limited** by **Thota Venkata Rao**. It is one of the companies approved for providing meaningful industrial exposure as per the guidelines of the All India Council for Technical Education (AICTE) and Andhra Pradesh State Council of Higher Education (APSCHE). An intern needs to do his/her internship in AICTE or APSCHE recognized companies only.

This internship has paved a way to plan for a few learning objectives that include,

- Get exposed to professional tasks and job roles in a reputed software organization and gain hands-on experience.
- Develop both technical and communication skills in the chosen field of work.
- Apply theoretical knowledge from coursework to real-world projects and scenarios.
- Enhance technical, analytical, and soft skills through active involvement in live projects.
- Cultivate problem-solving and critical thinking abilities to address challenges encountered during the internship.

The following are the learning outcomes, which include:

- The internship provided real-time exposure to software development tasks and enhanced practical knowledge.
- Helped in improving effective communication and in preparing concise technical reports.
- Successfully applied academic concepts to solve real-world problems in the company environment.
- Strengthened technical skills in programming and software tools, while also improving teamwork, time management, and interpersonal skills.
- Gained confidence in handling project challenges and identifying efficient solutions.

Overall, the internship at Sandspace Technologies has been a highly enriching experience that contributed to both technical and personal growth. It provided a platform to explore advanced software development practices, real-time project handling, and organizational culture. This internship not only improved problem-solving and teamwork skills but also gave exposure to cutting-edge, real-time projects, which will be valuable in selecting and executing future academic or professional projects.

CHAPTER 2: OVERVIEW OF THE ORGANIZATION

Sandspace Technologies Private Limited is a distinguished and successful software company based in Vijayawada, Andhra Pradesh, with a legacy of over 16 years in the software industry. The company has consistently demonstrated leadership in developing impactful, real-time projects that contribute to national development and technological advancement. With an annual turnover of ₹2.1 crore, Sandspace has established strong financial stability and sustained profitability, reflecting its commitment to innovation, quality, and sustainable growth.

Training / Internship Areas:

Sandspace provides opportunities for learning and professional development through its comprehensive internship and training programs in various domains, including:

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Web Development
- Cloud Computing
- Software Development and Testing
- Data Analytics
- Full Stack Java
- Mobile App Development

Training / Internship Methods:

Sandspace emphasizes hands-on learning by directly involving interns in real-time projects. The internship program follows a practical approach where students gain exposure to live projects under the mentorship of experienced professionals. Along with technical guidance, the company also provides access to tools, learning platforms, and industry-level practices.

Faculty / Mentors:

Interns are assigned dedicated mentors from the professional team who provide:

- Technical guidance and project specifications
- Regular progress reviews and feedback
- Code reviews and constructive suggestions
- Career development insights and skill-building support

Placement / Career Opportunities:

Sandspace uses its internship program as a talent pipeline, encouraging high-performing interns to explore professional roles within the organization. Its strong industry presence and focus on impactful projects help students gain practical skills and improve employability in the competitive job market.

Location:

Sandspace Technologies Private Limited is located in Vijayawada, Andhra Pradesh, India.

Vision:

The vision of Sandspace Technologies is to establish itself as a global leader in real-time software solutions, empowering national progress and transforming productivity in the technological landscape.

Mission:

The mission is centered on designing, developing, and delivering advanced software solutions that address critical national needs. The company is committed to technical excellence, innovation, and impactful contributions to both clients and society.

Core Values:

- Innovation
- Excellence
- Integrity
- National Impact
- Productivity Enhancement
- Collaboration & Teamwork
- Customer-Centric Approach
- Continuous Learning

Services Provided by the Organization:

- Real-time Software Development
- AI and ML-based Applications
- Web and Cloud Solutions
- Data Analytics and Business Intelligence
- Customized Project Development for National and Industrial Needs

Strategy:

- **Quality Delivery:** Focus on producing reliable and high-quality software solutions.
- **Innovation:** Encouraging creativity and fresh ideas to solve real-world challenges.
- **Hands-on Experience:** Emphasizing practical project involvement for interns and employees.
- **Collaboration:** Building strong teamwork and client partnerships.
- **National Development:** Aligning projects with the vision of contributing to India's digital growth.

Structure of the Organization:

Sandspace follows a hybrid organizational model, which includes both functional departments and project-based teams. The leadership structure consists of:

- Chief Executive Officer (CEO)
- Chief Technology Officer (CTO)
- Chief Operating Officer (COO)
- Departments: Software Development, Quality Assurance, Project Management, Business Development, HR & Administration, and IT Infrastructure Support.

Roles and Responsibilities:

- Developing and delivering innovative software solutions
- Conducting real-time project work
- Mentoring and guiding interns
- Providing technical and analytical support to clients
- Fostering teamwork and knowledge sharing

Future Plans of the Organization:

Sandspace Technologies plans to expand its research and development division, explore new service domains, and enter emerging global markets. The company aims to invest in cutting-edge infrastructure, strengthen employee training, and continue developing impactful projects in AI, ML, and other emerging technologies. With its strong vision, Sandspace aspires to maintain its leadership in the software industry while contributing to both national and global technological growth.

CHAPTER 3: INTERNSHIP PART

Description of Activities/Responsibilities

This chapter describes my internship journey at Sandspace Technologies Private Limited, where I worked as a Full Stack Java Intern for two months. My project during this internship was the development of a Library Management System (LMS), which combined frontend, backend, and database technologies to deliver a real-time, role-based web application.

The internship was structured to give exposure to software development practices, project workflows, debugging, testing, and deployment. I worked under the supervision of my mentor, who guided me in applying theoretical knowledge into practical, real-world development.

Working Conditions

- Internship conducted remotely with access to required tools and software.
- Mentor-led guidance through online meetings and code review sessions.
- Collaborative discussions to resolve project challenges.
- Weekly progress reporting and feedback to improve deliverables.

Weekly Work Schedule

- I worked around 49 hours per week.
- This schedule ensured a balance of learning new concepts and practical implementation of project tasks.

Technical Environment

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 500 GB
- Speed: 2.4 GHz or higher

Software Requirements

- **Operating System:** Windows 10/11
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Java, Spring Boot, Maven
- **Database:** MySQL, Spring Data JPA
- **Testing Tools:** Postman (API testing), MySQL Workbench
- **IDE:** IntelliJ IDEA / Eclipse

Project Title

"Library Management System (LMS) using Full Stack Java"

Project Overview

The Library Management System (LMS) is a web-based application designed to streamline library operations, including book management, user management, and borrowing/returning transactions. The system provides role-based access:

- **Admin** – Manage books, approve borrowing requests, view student/teacher records.
- **Student/Teacher** – Browse available books, request to borrow, return books.

This project replicates real-time industry development by integrating frontend design, backend logic, and database storage.

Project Objectives

- To develop a scalable and responsive web application.
- To implement role-based access with distinct features for Admin, Student, and Teacher.
- To integrate frontend (HTML, CSS, JS) with backend (Java, Spring Boot APIs).
- To design and manage a MySQL database schema for books and users.
- To test and validate APIs using Postman.
- To gain exposure to real-time project workflow, debugging, and deployment.

Project Approach & Detailed Tasks

1. Frontend Development (Client-Side)

- Designed user-friendly pages (login, dashboard, book listing) using HTML & CSS.
- Implemented JavaScript functions for interactivity (form validation, dynamic updates).
- Created a responsive layout to ensure compatibility across devices.

2. Backend Development (Server-Side)

- Developed RESTful APIs using Java Spring Boot.
- Used Maven for dependency management and build automation.
- Implemented Controller-Service-Repository architecture for clean code structure.
- Added authentication & authorization to differentiate between Admin, Teacher, and Student.

3. Database Integration (Persistence Layer)

- Designed database tables for:
    - Users (Admin, Teacher, Student roles)

    - Books (BookID, Title, Author, Year, Semester, Availability)
    - BorrowRecords (IssuedTo, IssueDate, ReturnDate, Fine)
- Used Spring Data JPA for seamless integration with MySQL.
- Applied constraints (Primary Key, Foreign Key, NOT NULL) to ensure data integrity.

4. Testing & Debugging

- Tested APIs using Postman (GET, POST, PUT, DELETE).
- Verified data consistency between frontend UI and backend database.
- Debugged SQL queries and API response issues.

5. Features Implemented in LMS

- **Admin Functions:**
    - Add / Delete / Update Books
    - View Borrow Requests
    - Approve or Reject Book Borrowing
    - View all Students & Teachers with borrowing history
- **Student Functions:**
    - View available books (by year/semester)
    - Request to borrow books
    - Return borrowed books (with fine calculation after 15 days)
- **Teacher Functions:**
    - View and borrow books (no fines applied)
    - Return borrowed books

Implementation & Output

- Created a book table that displays Title, Author, Year, Semester, and Availability.
- Added "View Book" option that shows book details (cover image, description).
- Integrated fine calculation system:
    - Students fined ₹1/day after 15 days.
    - Teachers exempted from fines.
- Achieved smooth interaction between frontend forms, backend APIs, and database records.

Skills Acquired

Technical Skills

- Strong foundation in Java, Spring Boot, Maven for backend development.
- Hands-on practice with frontend technologies (HTML, CSS, JavaScript).
- Database design and management using MySQL.
- Practical understanding of REST API development and testing.
- Exposure to project deployment and debugging.

Soft Skills

- Improved communication and collaboration through regular updates.
- Developed time management skills by balancing learning and project tasks.
- Enhanced problem-solving by resolving real-world technical challenges.
- Gained professionalism and accountability in meeting deadlines.

Conclusion

The Library Management System internship project at Sandspace Technologies provided me with end-to-end exposure to software development. I learned how different layers—frontend, backend, and database—interact to form a complete web application.

This internship not only improved my technical foundation in Full Stack Java development but also strengthened my soft skills such as communication, teamwork, and time management. The project helped me gain confidence to handle real-time industry projects and prepared me to work effectively in professional software development environments.

CHAPTER - 4
ACTIVITY LOG FOR THE FIRST WEEK - 1

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 19/05/25	Aim to know about the 3-2 summer internship and organizations who offering internships, which was conducted meeting by our college.	The learning outcomes from the discussion is gaining the knowledge about values of internship and organizations.	
Day – 2 20/05/25	Aim to select the organization and domain, in this free hand is given to us to select.	The learning outcomes from the Day -2 is to select the organization and domain.	
Day – 3 21/05/25	Aim to enroll to the organization and domain which was selected by us.	The learning outcomes from the Day -3 is to enrollment.	
Day – 4 22/05/25	Aim to onboarding us to the organization and more in detail about the internship and need to upload lecture notes in google form created by our college.	The learning outcomes from the meeting conducted by college is onboarding us and in detail about internship and google form to know our status.	
Day – 5 23/05/25	Aim to know about the organization and facilities offered by it for better internship experience.	The learning outcomes from the discussion is gaining the knowledge about organization and facilities given by them.	
Day – 6 24/05/25	Aim to know about the domain-java and where it is using and why it is trending.	The learning outcomes is to know about java and running applications in it and why it is trending.	
Day – 7 25/05/25	Aim to clear doubts and need to upload lecture notes in google form.	The learning outcomes is to clarification of doubts and upload notes.	

WEEKLY REPORT

WEEK – 1 (From Dt 07/06/24 to Dt 13/06/24)

Objective of the Activity Done:

Detailed Report:

Day- 1:

Day- 2:

Day- 3:

Day- 4:

Day- 5:

Day- 6:

Day- 7:

Then we use to have doubt sessions for the clarifying the of doubts & we need to upload lecture notes in the google form which was created by our college to track our internship status.

ACTIVITY LOG FOR THE SECOND WEEK - 2

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 26/05/25	Aim to learn about basics of java and what is java full stack & college meeting.	The learning outcomes of the session is to learn basic concepts of and about java full stack & college meeting also held.	
Day – 2 27/05/25	Aim to learn about variables, Data types & Non – primitive data types.	The learning outcomes of the session is to learn about variables, data types & Non – primitive data types.	
Day – 3 28/05/25	Aim to learn about Java tutorial from Geeksforgeeks it was given us task to complete it.	The learning outcomes of the session is assigned a work that is to complete the java tutorial from Geeksforgeeks.	
Day – 4 29/05/25	Aim to learn about java oops concept Abstraction & Encapsulation.	The learning outcomes of the session is to learn about oops concept Abstraction & Encapsulation.	
Day – 5 30/05/25	Aim to attend the online meeting from the college R&C division.	The learning outcomes of the session is track our internship status and clarifying our doubts.	
Day – 6 31/05/25	Aim to learn about remaining oops concept Inheritance & Polymorphism.	The learning outcomes of the session is to learn remaining oops concept.	
Day – 7 01/06/25	Aim to clear doubts and need to upload lecture notes in google form.	The learning outcomes is to clarification of doubts and upload notes.	